

PLACEMUX PREP

MERN FULL STACK DEVELOPER ASSESSMENT

OVERVIEW

MERN Full Stack Developer builds complete features from database to UI, owning the entire feature lifecycle.

3 Progressive Tasks:

- Task 1 (Easy): Build simple todo app with MERN (4 hours)
- Task 2 (Medium): Build job marketplace feature (5 hours)
- Task 3 (Hard): Build complete learning feature with real-time (6 hours)

TASK 2: JOB MARKETPLACE FEATURE (Medium)

Duration: 5 hours | Difficulty: Intermediate

Problem Statement

Build complete job marketplace: companies post jobs, students apply, see match percentage.

Requirements

- Database: 5 collections (companies, jobs, students, applications, skills)
- Backend: 12+ endpoints (CRUD for jobs, applications, search, filter)
- Frontend: Job listing, job detail, apply form, my applications
- Matching: Calculate skill match percentage
- Search: Filter by location, salary, job type
- Authentication: Basic JWT auth
- Validation: Input validation on all endpoints

Database Schema

- jobs: title, description, company, salary, location, required_skills, posted_date
- students: name, email, skills array, levels
- applications: student, job, status, applied_date
- companies: name, description, location

Backend Endpoints

- POST /jobs - Create job (company only)
- GET /jobs - List jobs with filters
- GET /jobs/:id - Get job detail

- POST /applications - Apply for job
- GET /applications - List my applications
- PUT /applications/:id - Update application status

Frontend Pages

- Job Listing: Search, filter, skill match indicator
- Job Detail: Full description, apply button
- Apply Form: Submit application
- My Applications: List status of applications

Tech Stack

- Backend: Node.js, Express, MongoDB, Mongoose
- Frontend: React, React Router, Axios
- Auth: JWT
- Testing: Jest, Supertest, React Testing Library

Deliverables

- backend/ (complete API)
- frontend/ (complete UI)
- DATABASE_SCHEMA.md
- API_DOCUMENTATION.md
- tests/ (backend + frontend tests)

Evaluation Rubric

Criteria	Weight	Excellent	Good	Score
Feature Completeness	25%	All features working	Missing features	—
Database Design	20%	Proper schema, relationships	Poor design	—
API Quality	20%	Clean RESTful API	Inconsistent endpoints	—
Frontend UI	20%	Clean, responsive, intuitive	Basic UI	—
Testing	15%	Good test coverage	Minimal tests	—